Xuanjun Chen

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Summary

Xuanjun is a Ph.D. student at National Taiwan University, advised by Prof. Hung-yi Lee and Prof. Jyh-Shing Roger Jang. His research focuses on **Speech/Audio Processing** and **Large Language Models**, specifically audio understanding & generation, deepfake speech, and agentic LLMs. His work is supported by NVIDIA Academic Grant and IEEE SPS Scholarship. He has published over 20 peer-reviewed papers, **15 as first/co-first author**, across top ML/NLP (ICLR, COLM, TIST, ACL Findings) and speech (TASLP, ICASSP, INTERSPEECH, etc.) venues. His research received the Best Paper Award at ROCLING 2025 and Best Paper Nominations at IEEE ASRU 2025 and ISCSLP 2026.

Education

Ph.D. Candidate in Communication Engineering, **National Taiwan University**, GPA: 4.3/4.3 — Sep 2023 – Present
M.S. in Computer Science and Information Engineering (CSIE), **National Taiwan University**, GPA: 4.19/4.3 — Sep 2020 – Jan 2023
B.S. in CSIE, **National Taiwan University of Science and Technology**, GPA: 4.11/4.3 — Jun 2020

Research Experience

NVIDIA Academic Grant Program, Lead Student Researcher, Taipei Taiwan — Oct 2026 – Mar 2027 (expected)

- **Audio Generation.** Leading research on fine-grained text-to-audio model to enhance audio generation capabilities.

Shanda Group, Research Intern (Remote), Tokyo Japan — Apr 2026 – Jul 2026

- **Speech Generation.** Investigated whether LLM agents can automate reinforcement learning for Text-to-Speech models. [1]

National Taiwan University, Research Assistant (Ph.D. & M.S.), Taipei Taiwan — Oct 2020 – Present

- Published **15 first/co-first** peer-reviewed papers, with 5 more currently under review.
- **Speech and Audio Language Models.** Analyzed the post-training dynamics of large audio language models [2], authored a high-impact survey [7], and served as a core contributor to DeSTA2.5-Audio [3], Dynamic-SUPERB [4], and Codec-SUPERB [5] [6].
- **Retrieval and Agentic LLMs.** Pioneered a dual-loop agentic framework for optimizing benchmark generation [8]; improved retrieval via graph navigation [9] and delta planning [10], alongside a systematic RAG study for historical archives [11].
- **Audio Deepfakes.** Conducting research on speech anti-spoofing, building upon prior work on audio deepfake detection across various scenarios [12]–[21] and audio-visual learning [22]–[24], including a system ranked 3rd of 42 teams in ASVspoof 2021 (LA track) [25].

Research Highlights

Total citations **600+** *as of Oct 2026.*

- MetaBench-Harness: Unlocking End-to-End Optimization of Benchmark Harnesses — *ICLR '27 (Submitted)*
  TL;DR: *A dual-loop framework where a meta-harness refines the harnesses that evolve benchmarks hard enough for frontier LLMs.*
- Can LLM Agents Automate Reinforcement Learning for Text-to-Speech? — *ICASSP '27 (Submitted)*
  TL;DR: *An LLM agent reconstructs and improves an RL post-training recipe for zero-shot TTS, cutting failure cases by half.*
- What Does LALM Post-training Learn? Rethinking Audio Understanding and Reasoning Capabilities — *NAACL '27 (Submitted)*
  TL;DR: *Much of the post-training gain comes from fitting answer formats and textual cues, not from better audio understanding.*
- Only Ask What You Don't Know: Grounded Delta Planning for Efficient Multi-step RAG — *COLM 2026*
  TL;DR: *Plans only for missing information after a first retrieval; top multi-hop QA accuracy at 22–68% lower cost.*
- CodaRAG: Connecting the Dots with Associativity Inspired by Complementary Learning — *ACM TIST 2026*
  TL;DR: *Brain-inspired RAG that chains dispersed evidence via associative graph navigation; +7–10% recall on GraphRAG-Bench.*
- CodecFake+: Codec-Based Resynthesized Data as a Proxy for Detecting CodecFake Speech — *TASLP 2026*
  TL;DR: *Re-synthesized speech from 31 codecs plus a codec taxonomy as a training proxy for detecting 17 CoSG deepfake systems.*
- Towards Generalized Source Tracing for Codec-Based Deepfake Speech — *ASRU 2025* · Best Student Paper Nominee
  TL;DR: *SASTNet fuses Whisper semantics with Wav2vec2/AudioMAE acoustics to trace CodecFake speech back to its source codec.*
- DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model with Self-Generated Cross-Modal Alignment — *TASLP 2026* ‡
  TL;DR: *General-purpose LALM trained on 5M self-generated audio QA pairs; keeps the backbone LLM's language ability.*
- Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Spoken Language Models with 180 Tasks — *ICLR 2025* ‡
  TL;DR: *180 community-contributed tasks over speech, music, and audio; no current model handles all of them well.*
- Codec-SUPERB: An In-Depth Analysis of Sound Codec Models — *ACL Findings 2024* †
  TL;DR: *Benchmarks neural audio codecs on resynthesis quality and downstream speech, speaker, emotion, and sound tasks.*

First/co-first author unless marked † (co-second author) or ‡ (contributing author).

Selected Publications

* *equal contribution;* † *co-second author;* ‡ *contributing author*

Speech and Audio Language Models and Generation

[1] **X. Chen**, et al., "Can LLM Agents Automate Reinforcement Learning for Text-to-Speech?" *ICASSP 2027 (Submitted)*

[2] H.H. Chen*, **X. Chen***, et al., "What Does LALM Post-training Learn? Rethinking Audio Understanding and Reasoning Capabilities," *NAACL 2027 (Submitted)*

[3] K.H. Lu, Z. Chen, ... **X. Chen**‡, ... H. Lee, "DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model with Self-Generated Cross-Modal Alignment," *TASLP, 2026*

[4] C.Y. Huang, ... **X. Chen**‡, ... S. Watanabe, H. Lee, "Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks," *ICLR 2025*

[5] H. Wu*, ... **X. Chen**†, ... H. Lee, "Codec-SUPERB: An In-Depth Analysis of Sound Codec Models," *Findings of ACL 2024*

[6] H. Wu, **X. Chen**, et al., "Codec-SUPERB @ SLT 2024: A Lightweight Benchmark for Neural Codec Models," *SLT 2024*

[7] H. Wu, **X. Chen**, et al., "Towards Audio Language Modeling – An Overview," *Technical Report, Feb. 2024*

Retrieval and Agentic Large Language Models

[8] **X. Chen**, et al., "MetaBench-Harness: Unlocking End-to-End Optimization of Benchmark Harnesses," *ICLR '27 (Submitted)*

[9] C.Y. Li*, **X. Chen***, et al., "CodaRAG: Connecting the Dots with Associativity Inspired by Complementary Learning," *ACM TIST, 2026*

[10] W.C. Chou*, **X. Chen***, et al., "Only Ask What You Don't Know: Grounded Delta Planning for Efficient Multi-step RAG," *COLM 2026*

[11] C. Lin*, B.H. Feng*, **X. Chen***, et al., "A Preliminary Study of RAG for Taiwanese Historical Archives," *ROCLING 2025* · Best Paper Award

Audio Deepfakes: Neural Codec, Singing Voice, Adversarial Attack, Conventional TTS/VC, and Audio-Visual Scenarios

[12] **X. Chen**, et al., "Why Is Spectrogram ResNet Still Effective in Singing Voice Deepfake Detection?" *ICASSP '27 (Submitted)*

[13] Y.A. Hsieh*, **X. Chen***, et al., "On Bona Fide Audio Discrepancies in Singing Voice Deepfake Detection," *ICASSP '27 (Submitted)*

[14] **X. Chen***, et al., "CodecFake+: Codec-Based Resynthesized Data as a Proxy for Detecting CodecFake Speech," *TASLP, 2026*

[15] **X. Chen**, et al., "Mitigating Proxy-to-Wild Domain Gap in Deepfake Speech," *ISCSLP 2026* · Selected Outstanding Paper

[16] **X. Chen***, C.Y. Hu*, et al., "Joint Fullband-Subband Modeling for High-Resolution SingFake Detection," *INTERSPEECH 2026* · Long Oral

[17] **X. Chen**, et al., "How Does Instrumental Music Help SingFake Detection?" *ICASSP 2026*

[18] **X. Chen***, I.M. Lin*, et al., "Towards Generalized Source Tracing for Codec-Based Deepfake Speech," *ASRU 2025* · Best Student Paper Nominee

[19] **X. Chen**, et al., "Codec-Based Deepfake Source Tracing via Neural Audio Codec Taxonomy," *INTERSPEECH 2025*

[20] **X. Chen**, et al., "Singing Voice Graph Modeling for SingFake Detection," *INTERSPEECH 2024* · Oral

[21] **X. Chen**, et al., "Neural Codec-based Adversarial Sample Detection for Speaker Verification," *INTERSPEECH 2024*

[22] **X. Chen**, et al., "Localizing Audio-Visual Deepfakes via Hierarchical Boundary Modeling," *ISCSLP 2026*

[23] **X. Chen**, et al., "Multimodal Transformer Distillation for Audio-Visual Synchronization," *ICASSP 2024*

[24] **X. Chen**, et al., "Push-Pull: Characterizing the Adversarial Robustness for Audio-Visual Active Speaker Detection," *SLT 2022*

[25] Y.L. Liao*, **X. Chen***, et al., "Adversarial Speaker Distillation for Countermeasure Model on Automatic Speaker Verification," *SPSC 2022*

Selected Honors, Service & Skills

Honors

- **NVIDIA Academic Grant Recipient** GPU compute resources via competitive selection by NVIDIA — 2026
- **IEEE SPS Scholarship** One of dozens of recipients worldwide from the IEEE Signal Processing Society — 2026
- **Google Student Travel Grant** Sponsored by Google Asia Pacific to present at ICASSP 2024 — 2024
- **CTCI Foundation Sci. & Tech. Scholarship** Bursary Award 2024 & Research Scholarship 2025 — 2024–2025
- **Dean's List** Certificate of Achievement for the top 5% of CSIE students, Taiwan Tech — Fall '18, Fall '19, Spr. '20

Service

- **Co-Organizer,** Responsible Speech and Audio Generative AI, Special Session at IEEE ASRU 2025 — 2025
- **Technical Committee,** Codec Speech Proc. Universal PERformance Benchmark Challenge, IEEE SLT 2024 — 2024
- **Reviewer:** AAAI, ICLR, NeurIPS, ICML; ACL, EMNLP, NAACL, COLM; ICASSP, INTERSPEECH, ASRU, SLT, etc. — 2023–Now

Skills

- **Coding & Toolkit:** Python, LaTeX; PyTorch, PyTorch Lightning, Hugging Face Transformers